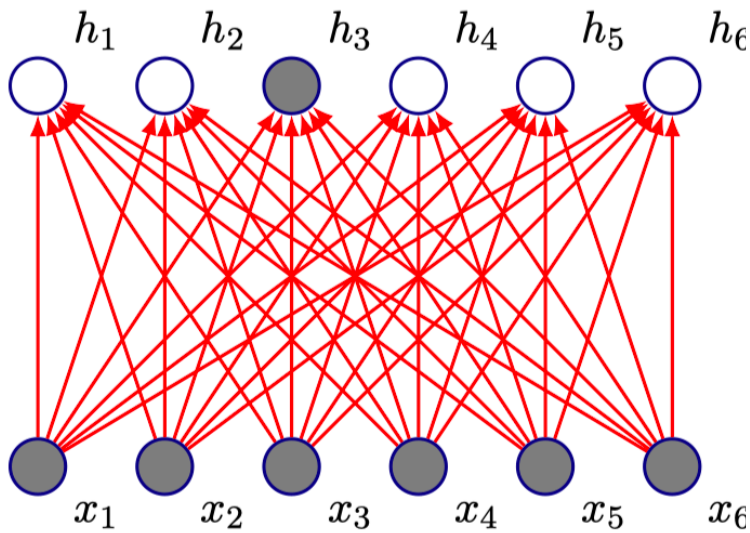


CNN Motivation

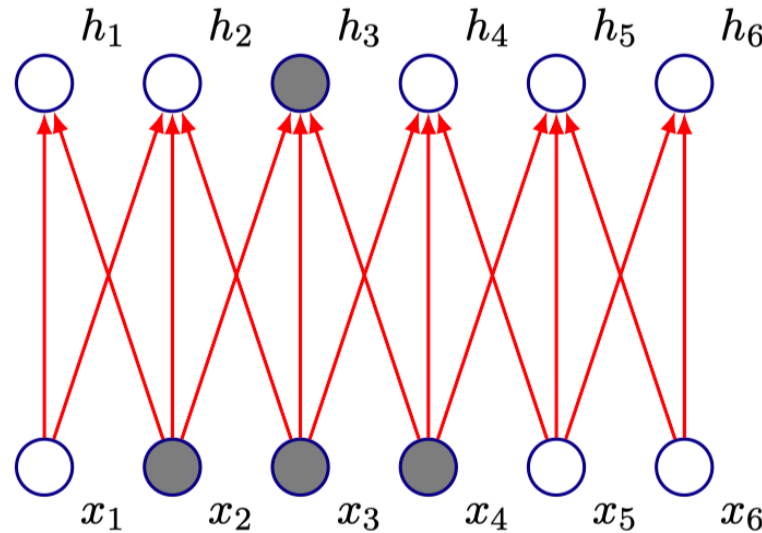
- ❑ Why not deep MLP (or feed-forward neural network)?
 - ❑ Deep multilayer perceptrons are **computationally expensive**, and **hard to train**.
 - ❑ Long training time, slow convergence, local minima
- ❑ Motivations of convolution
 - ❑ Sparse interactions
 - ❑ Parameter sharing
 - ❑ Translational equivalence
- ❑ The properties of CNNs are well aligned with properties of many forms of data (e.g. images, text), making them very successful

CNN Motivation: Sparse Interactions

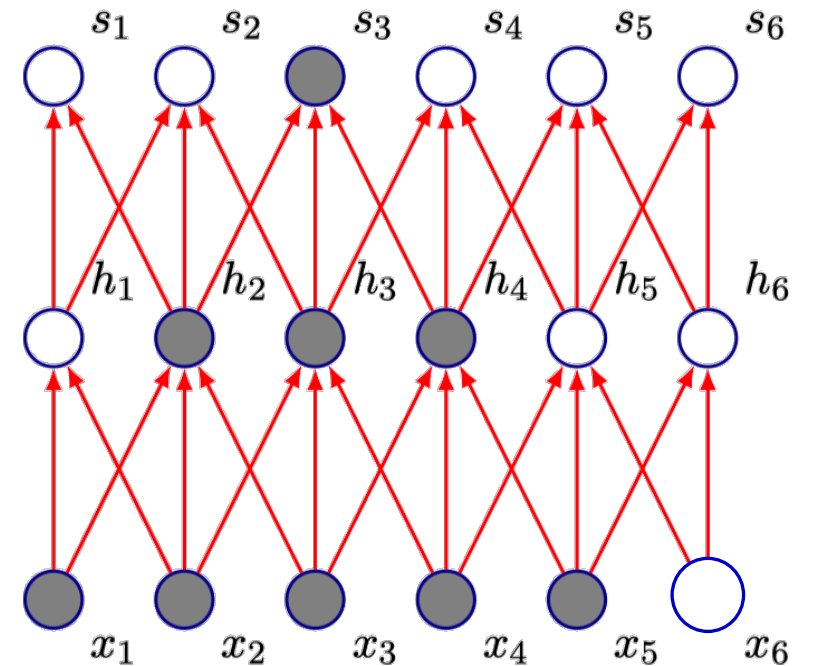
- E.g. 1D convolution with kernel size 3
- Units in deeper layers still connect to a wide range of inputs



MLP

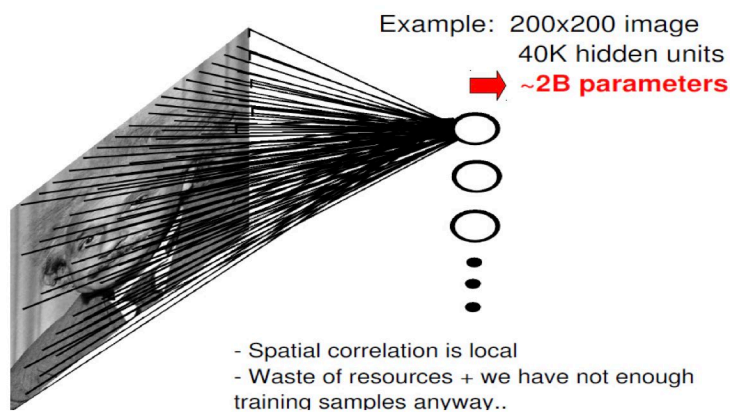
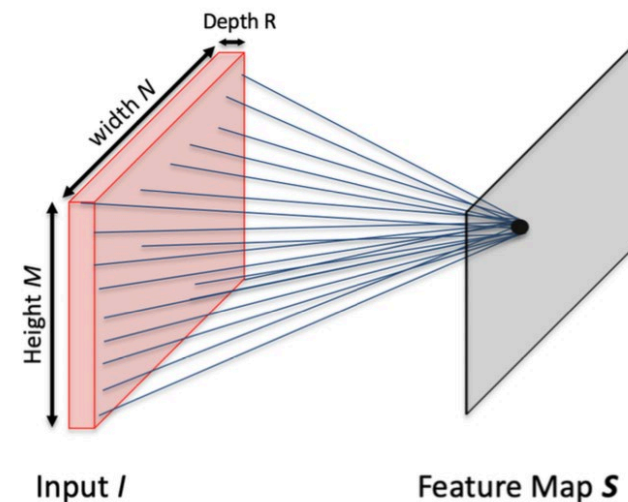


CNN

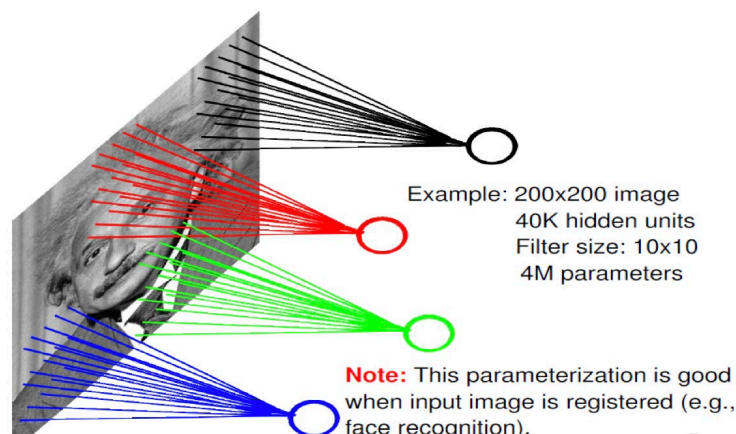


CNN Motivation: Parameter Sharing

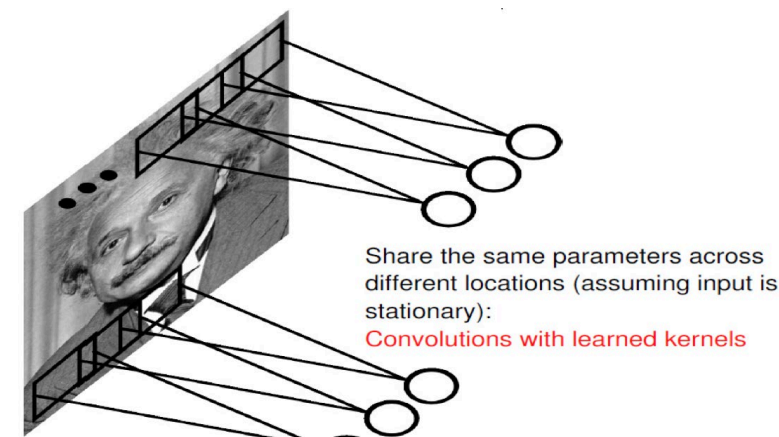
- ❑ Each kernel is used on all locations of input
- ❑ Reduce #parameters
- ❑ An Example
 - ❑ Given image of size $320 \times 280 \times 3$,
 - ❑ Generate a feature map of size 320×280 .
 - ❑ Option #1: use a kernel of size $5 \times 5 \times 3 \rightarrow 5 \times 5 \times 3 = 75$ parameters.
 - ❑ Option #2: use fully-connected layer: $320 \times 280 \times 3 \times 320 \times 280 = 24,084,480,000$ (more than 24 billion!) parameters.



Fully connected layer



Locally connected (sparse interaction)



Convolution (sparse + sharing)